

Exploring Deep Learning Approaches for Mammography Diagnostics with Down-Sampled and Cropped Images

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CERTIFICATE

This is to certify that the research work embodied in the dissertation entitled "**Exploring Deep Learning Approaches for Mammography Diagnostics with Down-Sampled and Cropped Images**" has been carried out at the School of Computational and Integrative Sciences, Jawaharlal Nehru University, New Delhi for the partial fulfilment of the requirement for the award of the degree of Master of Science in Computational and Integrative Sciences.

The dissertation embodies research work carried out by **Ankesh Kumar Pandey**, under the guidance of **Prof. Shandar Ahmad**, and has not been submitted in part for the award of any other degree or diploma to the best of our knowledge.

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My parents have been my greatest source of strength; this work is dedicated to them. I chose to work with an Indian-origin mammography dataset because my mother had breast cancer (CA Breast) and was diagnosed early, thanks to the non-invasive technique of Mammography. This personal experience inspired me to contribute to the field of cancer diagnostics in the way I could.

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Abstract

This project was started in the III semester of my coursework and it is being continued in this semester as well. It investigates the impact of reducing resolution and cropping mammography images on the performance of deep learning models for breast cancer classification.

It is being hypothesized that lower resolution images can minimize radiation exposure for patients while maintaining diagnostic accuracy. Various popular deep learning models will be evaluated to determine the minimum effective resolution for accurate classification of mammograms as cancerous or non-cancerous. Advanced techniques such as transfer learning and data augmentation will be leveraged to enhance model performance, addressing challenges related to limited dataset and variability in image quality.

Following the identification of the optimal model and resolution, super-resolution techniques will be employed to up sample the classified images. This step aims to improve visual interpretation for clinicians, enabling better diagnostic decisions without compromising patient safety. The research is expected to enhance the efficacy of mammography interpretation and contribute to the ongoing discourse on balancing diagnostic accuracy with patient care in radiology. The findings are anticipated to inform future practices in mammographic imaging and deep learning applications in medical diagnostics.

Keywords: Deep learning, medical diagnostics, convolutional neural networks, medical imaging, healthcare, artificial intelligence.

Introduction

Mammography is widely regarded as the gold standard for breast cancer detection [1] employing X-rays to produce detailed images of breast tissue. This screening method is crucial for identifying advanced as well as early-stage cancers, which can significantly improve treatment outcomes and reduce mortality rates. The American Cancer Society recommends annual mammography screenings for women aged 40 and older, emphasizing the importance of early detection in combating breast cancer. However, while mammography has proven to be an effective diagnostic tool, it is not without its drawbacks.

One of the primary concerns associated with mammography is the exposure to ionizing radiation. Each mammogram exposes patients to approximately 0.4 mSv of radiation, a dose equivalent to about 67 days of natural background radiation [2]. For individuals undergoing annual screenings, this radiation exposure can accumulate over time, raising concerns about potential long-term health risks, including an increased likelihood of developing radiation-induced cancers [3]. This issue is particularly pertinent for women who may undergo multiple screenings throughout their lives, highlighting the need for strategies that can mitigate radiation exposure without compromising diagnostic accuracy.

In light of these concerns, this project aims to explore innovative approaches that leverage deep learning techniques on low-resolution mammography images generated with reduced X-ray exposure [4]. The hypothesis proposes that it is possible to accurately predict breast cancer using these lower resolution images, thereby significantly diminishing the radiation burden on patients. By harnessing artificial neural networks, this research seeks to bridge the gap between effective cancer detection and patient safety.

The project encompasses these primary objectives-

First, an investigation and review of various image down-sampling techniques will be conducted to assess their applicability in mammogram classification [5]. This will involve analyzing how different down-sampling methods affect image quality and classification performance.

Second, a robust classifier model will be developed to differentiate between diseased and healthy tissue using the down-sampled data [4]. This model aims to maintain high levels of accuracy despite the reduced resolution of the input images.

Lastly, transfer learning of the model will be tested as well [6]. Through this research initiative, it is anticipated that advancements in imaging technology and artificial intelligence can enhance breast cancer screening protocols while prioritizing patient safety. By reducing unnecessary radiation exposure without sacrificing diagnostic accuracy, this project could contribute significantly to the ongoing discourse on improving mammography practices and ultimately lead to better health outcomes for patients undergoing breast cancer screening.

Background

Current mammography practices are primarily optimized for manual diagnosis, relying heavily on the expertise of radiologists to interpret the images produced. While traditional mammography systems have made significant advancements in image quality and dose optimization, they still face inherent limitations in detecting subtle details that may indicate the presence of breast cancer. The close similarity in radiation attenuation coefficients among various breast tissues can make it challenging for radiologists to distinguish between normal and pathological findings, particularly when the features of interest are on the order of micrometers [7]. Consequently, the accuracy of manual interpretations can vary significantly.

In contrast, deep learning models have shown great promise in enhancing diagnostic accuracy through automated image analysis. These models can process vast amounts of imaging data and identify complex patterns that may remain unnoticed by the human eye. By leveraging convolutional neural networks (CNNs) and other advanced algorithms, deep learning approaches can extract relevant features from mammographic images more effectively than traditional methods [8]. This capability not only enhances the detection of microcalcifications and other critical indicators of breast cancer but also allows for the analysis of lower resolution images without a significant loss in diagnostic performance [9].

The integration of deep learning techniques into mammography could revolutionize breast cancer detection by providing a more consistent and objective assessment of imaging data. By utilizing downsampled images generated with reduced X-ray exposure, this project aims to explore how these advanced computational methods can maintain or even improve classification accuracy while minimizing radiation risks associated with traditional mammography. This research is particularly relevant in light of recommendations from organizations such as the American Cancer Society, which advocate for regular screenings but also acknowledge the potential health risks associated with repeated X-ray exposure [10].

Ultimately, by harnessing the strengths of deep learning models, this project seeks to contribute to a safer and more effective approach to breast cancer screening, ensuring that patients receive accurate diagnoses while reducing their overall radiation burden.

Review of Literature

The integration of deep learning (DL) techniques into mammography represents a significant advancement in the field of breast cancer detection and diagnosis. Traditional mammography has long been the gold standard for screening, but it is primarily optimized for manual interpretation by radiologists. This reliance on human expertise can lead to variability in diagnostic accuracy, as subtle lesions and microcalcifications may be overlooked due to the limitations of human perception. Studies have shown that deep learning models can outperform human observers in detecting these critical details, providing a promising alternative to conventional methods.

Why are low resolution images necessary?

In X-ray imaging, as the pixel count increases (for example, when reducing pixel size to enhance spatial resolution), the area that each pixel covers decreases. Consequently, to maintain a consistent signal-to-noise ratio (SNR) and image quality, a greater total number of incident photons is required. This increase can indeed be approximated as growing with the cube of the pixel count because each pixel must receive a sufficient number of photons to accurately represent the image data. Specifically, if we halve the dimensions of each pixel (thus reducing the area by a factor of four), it would need to increase the total number of incident photons to achieve similar imaging performance [11].

On the other hand, the radiation dose is related to the total number of photons emitted during an X-ray exposure. When considering how dose relates to pixel count, if the pixel size is reduced (and thus increase pixel count), it may be needed to increase the total radiation output to ensure that each smaller pixel receives an adequate number of photons for accurate detection. This relationship can lead to a situation where decreasing pixel size results in a substantial increase in radiation dose, potentially increasing with the fourth power due to factors like increased exposure time and higher photon flux requirements [11].

Mammographic radiation exposure does affect patients undergoing mammography, it poses potential health risks, particularly the risk of radiation-induced carcinogenesis. Screening mammography is effective in detecting malignant diseases early, the use of ionizing radiation can be associated with a risk of radiation-induced breast cancer. This risk is particularly relevant given that mammography involves exposure to low-dose X-rays, which can contribute to the overall radiation burden on patients. It's important to note that the radiation dose is balanced as part of a trade-off with image quality [12].

Application of Deep Learning in Mammogram Classification

Mammogram classification has been greatly impacted by recent developments in deep learning, which have provided new strategies and architectures to improve diagnostic precision. A 2020 study by Salama et al [13] presented a sophisticated framework that combines several deep learning models (InceptionV3, DenseNet121, ResNet50, VGG16, and MobileNetV2) that were used to classify images from datasets like MIAS, DDSM, and CBIS-DDSM into benign and malignant. By utilizing both Cranio Caudal (CC) and Mediolateral Oblique (MLO) views for increased diagnostic precision, this framework also included a modified U-Net model for segmenting breast regions in the mammograms, with the goal of helping radiologists in early detection of disease. The study

effectively addressed the challenge of limited labeled data through transfer learning and data augmentation, achieving an impressive accuracy of 98.87% on the DDSM dataset.

In 2023, another study [14] shifted focus to improving breast cancer risk prediction in younger women by developing deep learning methods to estimate mammography density using low-dose mammograms taken at about one-tenth of the standard radiation dose. This research evaluated the quality and reliability of density scores derived from low-dose images and analyzed the impact of image resolution and training conditions on prediction performance. The researchers trained and tested their models on over 15,000 standard and simulated low-dose mammograms with known labels using two feature extraction techniques and an end-to-end training approach. They also evaluated performance using 296 corresponding standard and real low-dose images. The results indicated comparable prediction quality for both standard and low-dose images across all training conditions and resolutions. Notably, higher image resolutions improved performance for feature extraction methods, while the trained models consistently performed well across different resolutions. Furthermore, the study revealed a strong correlation between predictions for standard and low-dose images, with Spearman rank correlation coefficients of 0.951 at low resolution and 0.956 at high resolution, highlighting the reliability of the approach.

A 2017 study [15] proposes a computational approach for breast cancer modeling known as the Mammography-Histology-Phenotype-Linking-Model. This model claims to create a mapping of features and phenotypes that connect mammography abnormalities with their histopathological representations. The study discusses various challenges associated with this approach and highlights its potential contributions to clinical decision-making and treatment management. By linking imaging data with histological findings, the model seeks to enhance diagnostic accuracy and improve patient outcomes in breast cancer care. Prior research has employed deep learning techniques to classify mammograms as either healthy or diseased, as well as to categorize them into various BI-RADS classifications. However, only a limited number of studies have investigated the optimal image size for achieving the best disease detection outcomes.

Use of super-resolution techniques

Super-resolution is a powerful technique that creates high-resolution images from low-resolution ones, enhancing image quality without needing to upgrade existing imaging hardware. This approach has significant potential in medical imaging. A study by Ume-hara et al. [16] explored this concept using a Super-Resolution Convolutional Neural Network (SRCNN) to improve the resolution of digital mammograms. The SRCNN was trained to reconstruct high-resolution images from low-resolution inputs, and its performance was compared to traditional linear interpolation methods, such as nearest neighbor and bilinear interpolation.

The study examined how well SRCNN-enhanced images correlated with clinical features of mammography lesions, including breast density, BI-RADS assessments, and pathology-confirmed diagnoses. Image quality was assessed quantitatively using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). The findings revealed that SRCNN significantly outperformed linear interpolation methods in image quality ($p < 0.001$). Notably, SRCNN-enhanced images showed greater quality improvements in cases of dense breasts, high-risk BI-RADS categories, and pathology-confirmed malignant lesions compared to less dense, lower-risk, or benign cases ($p < 0.01$). The study concluded that SRCNN is an effective tool for enhancing digital mammograms,

particularly for magnified images in more complex scenarios.

Breast cancer remains the leading cause of cancer-related deaths among women globally, according to the World Cancer Research Fund. Early detection significantly improves survival rates, making high-quality mammography vital for effective screening. However, achieving higher image resolution often requires increased radiation exposure, which poses health risks. To address this challenge, researchers have developed methods to enhance mammogram resolution without raising radiation doses. Recent advancements in computer vision have focused on improving image resolution in areas like facial imaging. Inspired by these developments, Zheng J. et al. [17] adapted such techniques for medical imaging, particularly mammograms. Their study introduced an end-to-end method that generates high-resolution mammograms from low-resolution inputs created with minimal X-ray exposure. This approach uses a mesh warping algorithm for automatic image alignment, followed by a two-step super-resolution process combining a PCA-based global model with a patch-based local model. Experimental results demonstrated that this method effectively produces high-quality mammograms from low-resolution images without requiring manual adjustments, offering a promising solution for safer and more efficient breast cancer screening.

Deep learning models capture minute details in low resolution images

Current radiological practices often prioritize images that are visually interpretable by clinicians, potentially overlooking the superior pattern recognition capabilities of machines. This approach can result in unnecessarily high-dimensional images and increased radiation exposure, which has not been adequately addressed in image analysis studies. In a study conducted by Vinayak N. et al [18], they have concluded that a CNN model like EfficientNetB0 is very promising in predicting diseased versus healthy condition by using low resolution diagnostic images like chest X-ray image. In this study the researchers tried to do a classification between COVID-19, Pneumonia and healthy individuals by looking at their chest X-ray images using CNN models. Existing research on detecting COVID-19 and pneumonia using chest X-ray images typically involves reducing the original image size to various lower dimensions, often around 224×224 pixels. This study aims to investigate the impact of image size on model performance by examining sizes ranging from 300×300 pixels down to a minimal 1×1 pixel.

The findings indicate that as image size decreases, the visible information to the human eye also diminishes; however, neural networks can still effectively process lower-dimensional images. This raises two key questions: first, which X-ray dimensions are optimal for effective disease detection? Second, if lower-dimensional images can accurately identify diseases, can X-rays be collected at these reduced dimensions to minimize patient radiation exposure? The study emphasizes the trade-off between radiation exposure and detection performance across different image sizes. The study investigates three binary classification problems: Covid-19 vs. Normal, Covid-19 vs. Pneumonia, and Pneumonia vs. Normal. Conclusively, it can be said that the deep learning models can effectively learn from low-dimensional chest X-ray images, where human interpretation may fall short. By leveraging advancements in artificial intelligence, it is possible to capture smaller X-ray images, significantly reducing radiation exposure for patients. To create higher-dimensional images for medical practitioners, generative models such as Generative Adversarial Networks (GANs) can be used. This approach benefits both

patients and doctors by minimizing radiation exposure while providing high-quality scans for review.

Feature extraction-based approach for classification

The transformer architecture, introduced in the revolutionary paper “Attention is All You Need” [19], marked a significant departure from recurrence-based neural networks for sequence modeling tasks. The key innovation of transformers lies in their self-attention mechanism, which allows the model to weigh the importance of different elements in a sequence when making predictions. The self-attention mechanism can be described as mapping a query and a set of key-value pairs to an output, this enables the model to focus on relevant parts of the input sequence regardless of their position.

The Vision Transformer (ViT) [20] adapts this architecture to image analysis by treating an image as a sequence of patches. The process involves, (i) splitting an image into fixed-size patches, (ii) linearly embedding each patch, (iii) adding position embeddings, (iv) feeding the resulting sequence of vectors to a standard Transformer encoder and (v) Using a classification token (CLS) to obtain the final representation for classification. This approach allows the model to learn global relationships between different regions of the image without the inductive biases inherent in CNNs.

All of this literature has been studied and a prominent research gap was found out which was: “although deep learning techniques have and are being applied in image based diagnosis of breast cancer, it hasn’t considered the effect of low resolution images and the use of transformers for extracting latent space features.”

Objectives

Three objectives of this project were been defined-

- To investigate and review image down sampling techniques applied to mammogram classification.
- To develop a classifier model which classifies the down sampled data into diseased vs healthy.
- To attempt to build a model which is able to achieve transfer learning, and predicts across datasets.

The first objective encompasses majorly two areas, researching about the appropriate dataset and downloading it and then preprocessing the images. Preprocessing includes converting the images to workable file format to start with. Medical images are stored in DICOM (.dcm) file format as a standard, so converting these images to a lossless file format (like .png). Studying various different image downsampling and cropping techniques and applying them to the dataset, checking the redundancy in the dataset, compiling and cleaning the image metadata are some of the preprocessing which needs to be done.

The second objective includes exploring and studying various neural networks, particularly convolutional neural networks (CNNs), as these are the neural networks specifically designed and used for image classification. The neural networks once chosen to work with can be then applied on our dataset, at various different image resolutions,

then their performance can be recorded to find the optimum image resolution which is good for predicting the disease on the one hand and is able to save the patients from unnecessary X-ray exposure as well.

The third objective tries to learn about whether the model which is being developed works across datasets or not? Whether the performance which is being recorded by training and testing on datasets of one demography works on datasets of other demography or not? These are the research questions which will be dealt under this objective.

Methods

A comprehensive work plan was decided upon to start working on this project. Each part of the plan is being explained here in detail in the figure 1.

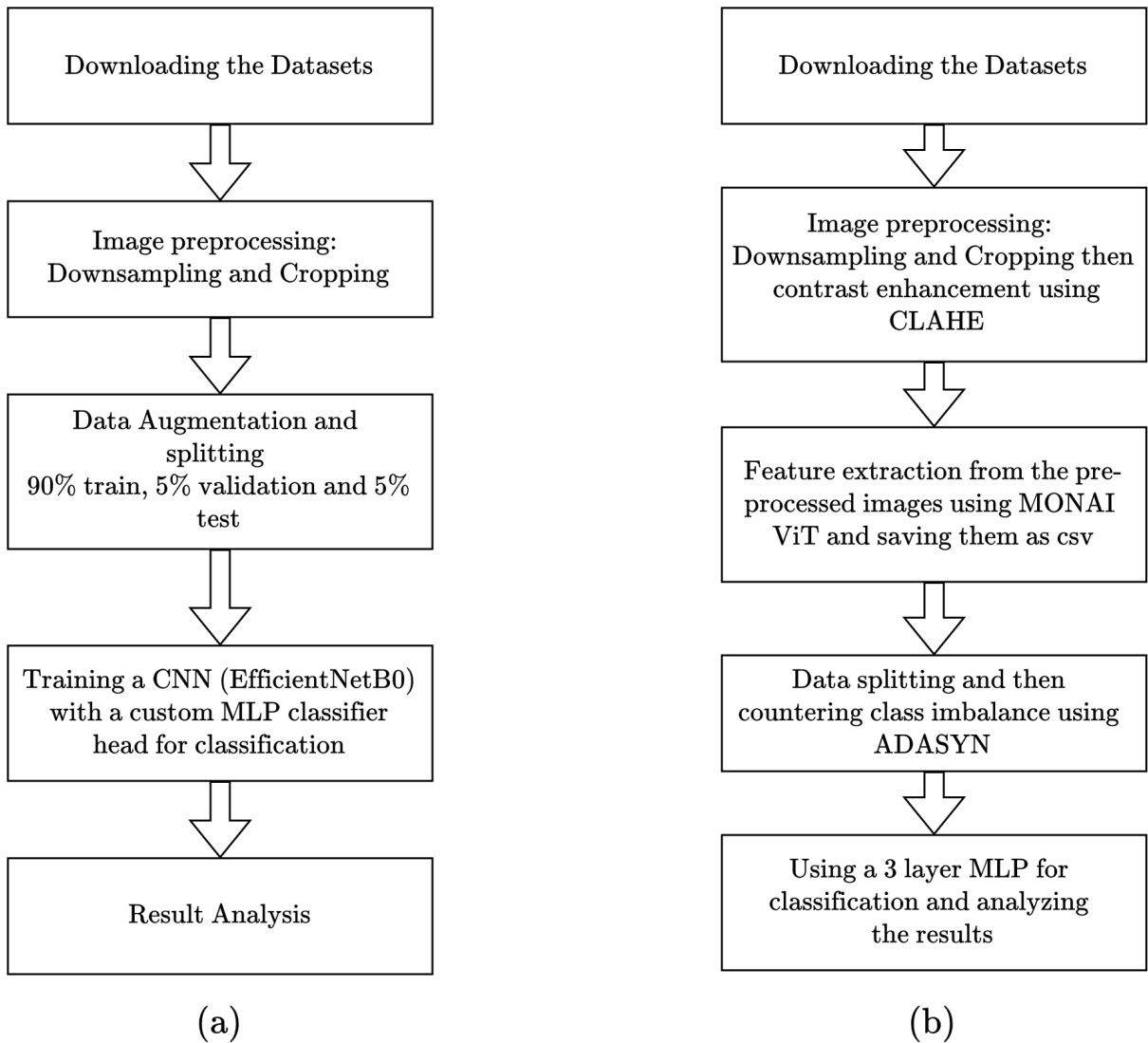


Figure 1: (a) Workflow of the CNN based training and classification (b) Workflow of the ViT feature extraction and classification

Datasets

The MAMOS_1000000004 Dataset - It is a mammography dataset coming from the Indian population. It was generated at All India Institute of Medical Sciences, New Delhi under the project titled ‘An Opportunistic screening mammography dataset from a screening-naive population’ [21]. This dataset is hosted on the Indian Biological Images Archive website, a unit of Indian Biological Data Centre (IBDC). It is an open-source dataset containing 3577 images, in which 1869 Full Field Digital Mammography & 1708 Synthetic Mammography images are there. Images in this dataset are in DICOM (.dcm) format. It provides breast-level imaging data (BIRADS category and breast density) along with ground truth labels based on histopathology for cancers and follow-up scans for non cancers.

This dataset was chosen for this mini project due to 3 major factors-

- This dataset has been created in India, thus making it more relevant for a study being conducted in this country.
- All the other credible datasets such as those which are discussed previously, were generated fairly long ago, and the MAMOS_1000000004 Dataset was released in September 2024, making it a novel data to work with.
- The size of this dataset is approximately 67 GB, making it much more manageable as compared to datasets like CBIS-DDSM, sized around 163 GB.

The Chinese Mammography Dataset (CMMD) - Another dataset of Chinese origin, published in 2021 was acquired from “The Cancer Imaging Archive (TCIA)” [22]. It had a total of 2214 full-field digital mammograms, in which 1084 images were of benign and 1130 images were of malignant origin.

Image Preprocessing & Down-sampling

After we had acquired the datasets, certain pre-processing steps consistent with the published literature were performed, which are being described in subsequent paragraphs.

Images had to be converted from their native DICOM format to PNG. DICOM stands for Digital Imaging and Communications in Medicine, a standard format for storing, transmitting, and displaying medical images and related data, it was developed by jointly by National Electrical Manufacturers Association (NEMA) and the American College of Radiology (ACR). DICOM is used as the format of choice for storing medical images mainly because of the metadata which it carries. It has around 300 metadata attributes such as patient information, study information, modality (e.g. CT, MRI, Mammography etc), image information (e.g. image position, orientation, pixel array, pixel spacing etc), equipment information, acquisition parameters (e.g. kilovoltage peak, exposure time, tube current etc). While DICOM file format is great for data storage it is not that good for working with the data, as due to it’s metadata the individual file size becomes hefty, due to these constraints the images were converted in a lossless file format .png. This helped in reducing the file size by a factor of 10 while preserving the details of the image. After conversion the dataset’s size was 6.7 GB.

Once all the images were converted, they were cropped to further reduce the redundancy in them, as the black region in a mammogram acquires 60-90% of the area it is

imperative to remove the black region from the image as much as possible, while preserving all of the tissue area. It makes the computation much more efficient. For cropping many techniques were studied, namely-

- Intensity based thresholding- Very high efficiency, moderate accuracy
- Contour detection- High efficiency, moderate to good accuracy
- Convex hull- Moderate efficiency, high accuracy
- Deep learning segmentation- Moderate to low efficiency, very high accuracy.

Based on the advantages and the drawbacks offered by these methods, Convex Hull was found to be the optimum method for cropping the images. This algorithm finds the smallest convex shape that encloses a set of points in a plane or space. In the context of breast tissue segmentation, it helps define the contour of the breast region and exclude surrounding non-tissue area by wrapping the breast tissue outline. OpenCV was used to implement this cropping onto images.

Indian Biological Images Archive provides other details of the dataset in formats like .xlsx (excel spreadsheet) and .csv (comma separated value file). Thus, the accompanying data was downloaded as three distinct .csv files, which contained sample id, experiment id, BIRADS categories, ground truth label and other features. In this stage of preprocessing the most important features of the dataset were extracted from these three files and compiled into one file. The final file contained sample id, image file name, and ground truth label. This file was further used as the file which contained the labels of all the images in the dataset.

Sample-wise Normalization

It was observed that the images were having widely varying pixel intensity distributions, which is a common trait in medical images. Sample wise normalization of ImageDataGenerator functionality of Keras API was used to do the z-score normalization. The images were first rescaled to have a pixel intensity between 0 to 1 by dividing the pixel values by 255. Then z-score normalization was done on the rescaled images. This helped the model in focusing on structural patterns rather than absolute intensity values. Unnormalized images may introduce biases due to different contrast levels, brightness variations, or scanner-specific characteristics. Sample-wise normalization helps the model generalize better across different datasets and imaging conditions.

The CMMD dataset also underwent the exact same pre-processing steps discussed in preceding sections, but was kept reserved as an external testing data for the trained models. This dataset was solely used to test the “Transfer Learning” of features across datasets of different geographic origin.

Deep Learning

This has been the most crucial and resource intensive step. Various DL models were studied and it was decided that the deep learning step will start with the model EfficientNet B0. EfficientNet-B0 is the baseline model in the EfficientNet family of convolutional neural networks (CNNs), developed by Google Research in 2019 [23]. This architecture strikes a balance between high accuracy and computational efficiency, making it ideal for applications ranging from mobile and edge computing to image processing tasks.

Key Features of EfficientNet-B0

- **Compound Scaling:** EfficientNet-B0 uses a unique compound scaling method that scales the network’s depth, width, and resolution uniformly based on fixed coefficients. This allows the model to adapt efficiently to varying computational resources while maintaining strong performance across tasks.

Architectural Highlights

- **MBConv Blocks:** The backbone of EfficientNet-B0 consists of Mobile Inverted Bottleneck Convolution (MBConv) blocks. These blocks improve performance with depthwise separable convolutions and squeeze-and-excitation (SE) mechanisms, enhancing the network’s ability to represent features effectively.
- **Layer Structure:** The model has 237 layers organized into seven blocks, each configured to optimize learning and performance.
- **Parameters:** For this specific implementation, the model contains 5,330,571 parameters, of which 5,288,000 are trainable and 42,571 are non-trainable.

Performance: EfficientNet-B0 achieves an impressive top-1 accuracy of 77.1% on the ImageNet dataset while using significantly fewer parameters compared to other models. This combination of accuracy and efficiency makes it particularly well-suited for deployment in environments with limited computational resources.

Dealing with class imbalance in the data

After cleaning the dataset, and the file containing all the labels for the images it found that there is severe class imbalance in the dataset. Here is the table 1 summarizing the imbalance:

Table 1: Distribution of dataset images.

Class	Number of Images
Cancer images	299
Non-cancer images	3278
Total	3577

Data Splitting & Model Training

To deal with this imbalance in the data, data augmentation strategies were used for increasing the number of data points in the minority Malignant class. Class weights were also defined to penalise the model for classifying the majority class wrongly and reward the model for correctly classifying minority class. This discrepancy of the dataset has been a big hurdle while working with this data, in the progress report presented in the mini project report of III semester, it was mentioned that undersampling of the majority class was done to counter the class imbalance, in this semester a different approach of calculating and using the “Class weights” for training has been explored.

Data Augmentation

The dataset was artificially expanded by using python code by the ImageDataGenerator from Keras API, through the application of various transformations to the images in the Malignant class.

Augmentation Parameters:

- **Rotation range = 15:** The image was randomly rotated within a range of -15 to +15 degrees, enabling the model to become invariant to small input rotations.
- **Width shift range = 0.1:** Horizontal shifts of up to 10% of the image width were applied, simulating translations to increase robustness against slight shifts.
- **Height shift range = 0.1:** Vertical shifts of up to 10% of the image height were applied, enhancing robustness against vertical translations.
- **Shear range = 0.15:** A shearing transformation was applied, skewing the image along one axis, which is useful for simulating perspective distortions.
- **Zoom range = 0.11:** Random zooming in or out by up to 11% was applied, making the model robust to variations in scale.
- **Horizontal flip = True:** The image was flipped horizontally with a 50% probability, which is particularly useful for images where horizontal symmetry is meaningful (e.g., chest X-rays, mammograms).
- **Brightness:** Brightness of the images was varied in a range of 0.7 to 1.3.
- **Fill mode = 'nearest':** Missing areas caused by transformations such as rotation or shifting were filled using the nearest pixel values.

Processing & Augmenting the Images in minority class

The images in the Malignant class were first split into train, validation and test sets by specifying these directories. Each of these sets had two classes Benign and Malignant. The 'Malignant' images from the train directory were iterated through by the code. Each image was loaded and converted into a NumPy array for further processing. The array was reshaped to include a batch dimension $((1,) + \text{img_array.shape})$, as batch input is expected by the datagen.flow method. Augmented images were generated separately by the datagen.flow method, based on the specified augmentation parameters. The augmented images were then saved. For every original image, 11 augmented images were generated. This ensured the augmented dataset was balanced and not overly dominated by artificially generated samples. Here is the table 2 showing before and after statistics of the augmentation:

Table 2: Distribution of training images before and after data augmentation.

Training Images	Before Augmentation	After Augmentation
Malignant images	240	2640
Benign images	2621	2621
Total	2861	5261

No augmentation was applied on the validation and test set images to maintain the real life consistency in the images. Finally, both the validation and test sets had 30 images of malignant pathology and 328 of benign pathology, thus each having 358 images.

Results

The dataset finally had a split in 90:5:5 ratio, i.e. 90% of the images in the training set, 5% images in the validation set and 5% images in the test set. Once the data splitting was done, class imbalance was taken care of by defining doing augmentation before hand and class weighting on the fly. EfficientNet B0 was run on the dataset for 5 different resolutions 300×300 px, 224×224 px, 175×175 px, 128×128 px and 64×64 px. Early stopping criteria with a tolerance of 6 epochs was applied. ElasticNet regularization (L1 and L2) was applied with the L1, L2 values of 2.4396×10^{-5} and 2.9038×10^{-6} respectively. Train batch size was taken as 32 and validation batch size was taken as 16, to ensure better implementation of regularization. A learning rate scheduler was used which monitored validation accuracy and reduced the learning rate by a factor of 0.5 upon continued stagnation of validation accuracy for 6 epochs. Important statistics like training accuracy, training loss, validation accuracy, validation loss were recorded and plotted

CNN Results

Here is the table 3 summarizing all the statistics which were collected during the neural network run.

Table 3: Performance metrics across different input image resolutions.

Resolutions	Val. Accuracy	Val. Loss	Test Accuracy	Test Loss
300×300 pixels	0.92	0.08	0.92	0.09
224×224 pixels	0.87	0.19	0.83	0.17
175×175 pixels	0.93	0.12	0.92	0.15
128×128 pixels	0.91	0.11	0.91	0.10
64×64 pixels	0.86	0.15	0.85	0.13

In figure 2 we can see how the accuracy and loss vary with varying resolution of input images.

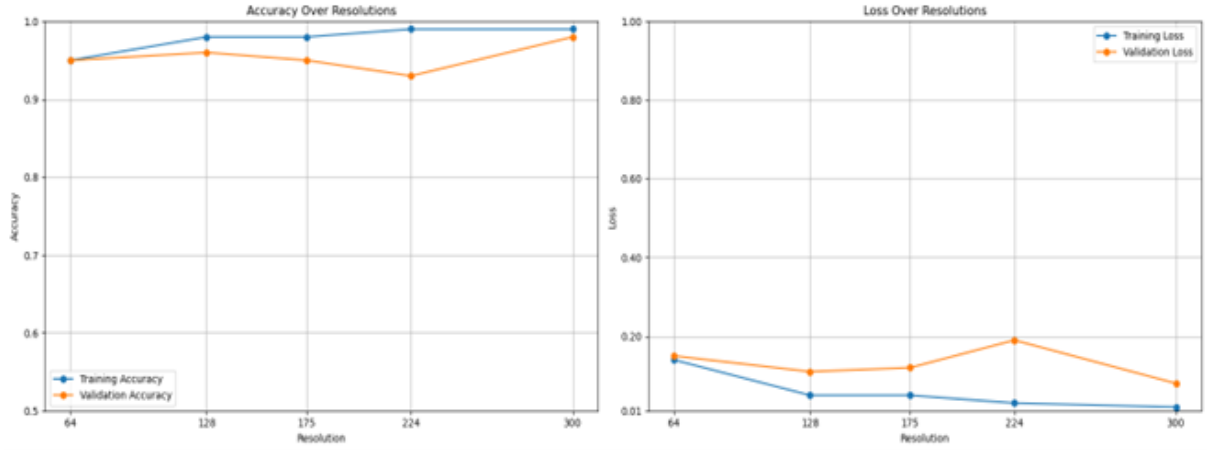


Figure 2: Accuracy and loss plots from the MLP training done to classify the images using EfficientNet-B0 features

These are the individual accuracy and loss curves against epochs plotted for the five different input resolution of the images. From the accuracy and loss plots we can see that as we move down from 300x300 pixels, we register a loss in accuracy, and gain in the loss of training and validation sets both, especially with the validation set. As far as the success of this particular CNN (EfficientNetB0) with our dataset and hyperparameters of choice is concerned we are concluding that an image of 175x175 pixels is good enough for EfficientNetB0 for classifying the Benign versus Malignant images with an accuracy of 0.92 (92%) while registering a loss of 0.15.

Vision Transformer based feature extraction

For this study, we implemented a ViT [20]-based approach for mammogram classification using the MONAI library and the PyTorch framework. MONAI stands for “Medical Open Network for AI” originally developed by NVIDIA and King’s College London. It is a collaborative initiative that brings together leading researchers, clinicians, and industry experts to advance medical imaging AI. The implementation follows these key steps:

Data preparation & preprocessing

Our mammography dataset contains 3,577 images, of which 3,277 are benign and 300 are malignant, representing a significant class imbalance, common in medical imaging datasets. Preprocessing steps included:

- Converting the images from their native DICOM (.dcm) format to .png for easy computation.
- Image normalization to standardize pixel values.
- Using Contrast Limited Adaptive Histogram Equalization.

Contrast Limited Adaptive Histogram Equalization (CLAHE): Contrast Limited Adaptive Histogram Equalization (CLAHE) [24], is a widely used image enhancement technique, particularly effective in the biomedical domain. CLAHE improves local contrast in images by applying histogram equalization to small regions (tiles) rather than

the entire image, and limits the amplification of noise by clipping the histogram at a predefined threshold. This method is especially beneficial for medical images, such as X-rays and retinal scans, where subtle contrast differences are crucial for accurate diagnosis. CLAHE has demonstrated superior performance over traditional methods, enhancing fine details while preserving overall image quality.

Model architecture

We utilized the pre-trained Vision Transformer model from MONAI, which is specifically optimized for medical imaging tasks. The utilized hyperparameters given in the table 4. The model architecture is adapted for single-channel grayscale inputs, structured into three primary stages:

Input and Patch Embedding

- **Input channels:** 1 (grayscale image)
- **Image size:** 224×224 pixels
- **Patch size:** 16×16 pixels
- **Patch embedding block:**
 - Two-dimensional convolution layer (Conv2d : $1 \rightarrow 768$) with a kernel size of 16×16 and a stride of 16 to project non-overlapping patches into the latent embedding dimension.
 - Dropout regularization with rate $p = 0.0$.

Transformer Encoder

The encoder consists of 12 identical sequential **TransformerBlock** layers. Each block contains the following sub-modules:

1. **LayerNorm 1:** Layer normalization applied prior to the self-attention mechanism (pre-norm configuration).
2. **Self-Attention Block (SABlock):**
 - Linear projection layer mapping embeddings to Query (Q), Key (K), and Value (V) representations ($768 \rightarrow 2304$).
 - Multi-head attention mechanism comprising 12 attention heads ($d_{\text{head}} = 64$).
 - Linear output projection layer (Linear : $768 \rightarrow 768$).
 - Tensor rearrangement operations for multi-head attention matrix computations.
 - Dropout applied to attention weights and output projections with $p = 0.0$.
3. **LayerNorm 2:** Layer normalization applied prior to the multilayer perceptron feedforward network.
4. **Multilayer Perceptron (MLP) Block:**

- First linear transformation (Linear : $768 \rightarrow 3072$) with Gaussian Error Linear Unit (GELU) activation.
- Second linear transformation (Linear : $3072 \rightarrow 768$).
- Dropout applied after each linear layer with probability $p = 0.0$.

Final LayerNorm

- A terminating normalization layer (LayerNorm : 768) applied to the latent representations emerging from the 12th transformer encoder layer.

Table 4: Summary of Vision Transformer structural hyperparameters.

Component / Hyperparameter	Specification
Input Image Dimensions	$1 \times 224 \times 224$
Patch Size	16×16
Number of Patches (N)	196
Embedding Dimension (D)	768
Transformer Encoder Layers	12
Attention Heads	12
Head Dimension (d_{head})	64
MLP Hidden Dimension	3072
Activation Function	GELU
Dropout Probability (p)	0.0

Using this model architecture features were extracted from the images, for classification separate classification models were used. After extracting the features, they were saved in a .csv file. The feature file had 3577 rows, for the same number of images and 767 columns for extracted features plus 1 column for class labels, 0 for benign and 1 for malignant. Once we had the feature and label dataset, it was decided to work on managing the class imbalance in the dataset.

Oversampling to improve class imbalance – ADASYN

ADASYN (Adaptive Synthetic Sampling) [25] is an advanced oversampling technique designed to address class imbalance in datasets, particularly improving the learning of minority classes. Unlike Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples uniformly for the minority class, ADASYN adaptively focuses on harder-to-learn samples by generating more synthetic data in regions where the minority class is sparsely represented. This targeted approach helps classifiers better distinguish minority instances, especially in complex or overlapping regions, leading to improved recall and overall model performance. Studies have shown that ADASYN can outperform SMOTE and other traditional methods, especially in challenging, highly imbalanced scenarios [26].

Here is the table 5 showing class distribution before and after using ADASYN:

Table 5: Dataset sample distribution before and after augmentation.

	Benign	Malignant	Total
Before	3277	300	3577
After	3277	2936	6213

After doing the data augmentation, we went forward for classifying the data with a simple multilayer perceptron.

Classification using MLP

We employed a simple MLP for classification. The MLP had 3 hidden layers with 512, 256, 128 neurons respectively in each layer, each having ReLU activation. The output layer had 1 neuron with sigmoid activation. The total no. of trainable parameters were 558,081. L1, L2 regularization was applied, with values: L1= 0.000002 and L2= 0.000141. Class weights and weighted accuracy were calculated. Training history, AUC-ROC, confusion matrix were plotted.

Here are the training curves, AUC-ROC obtained on the held out test set (from the MAMOS dataset) and the confusion matrix, shown in the figures 3 and 4.

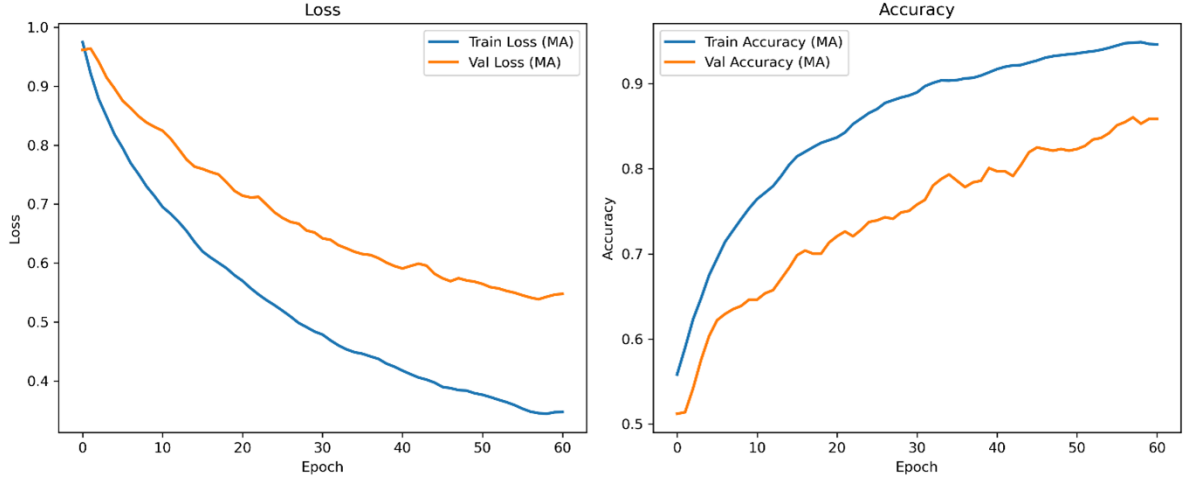


Figure 3: Training curves; training and validation accuracy and loss against training epochs

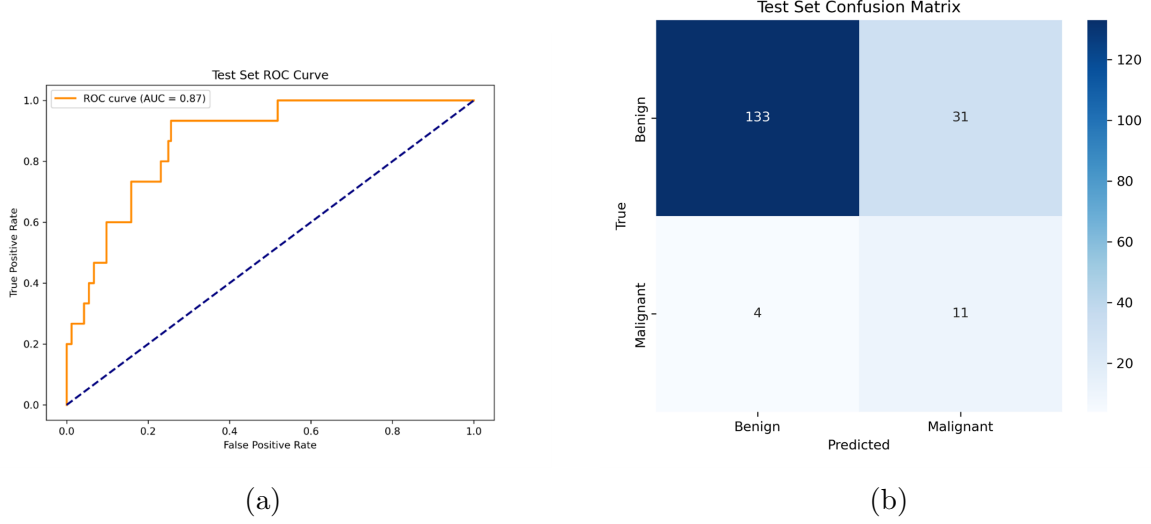


Figure 4: (a) AUC-ROC curve obtained after evaluation on the held out test set, and (b) Confusion matrix obtained after evaluating on the held out test set

Transfer Learning Results

In the last part of this dissertation, we tried testing the cross-dataset transferability of learning of the MLP with ViT features extracted from the CMMD dataset, after subjecting it to the same data pre-processing pipeline. The AUC-ROC curves of this experiment are shown in the figure 5.

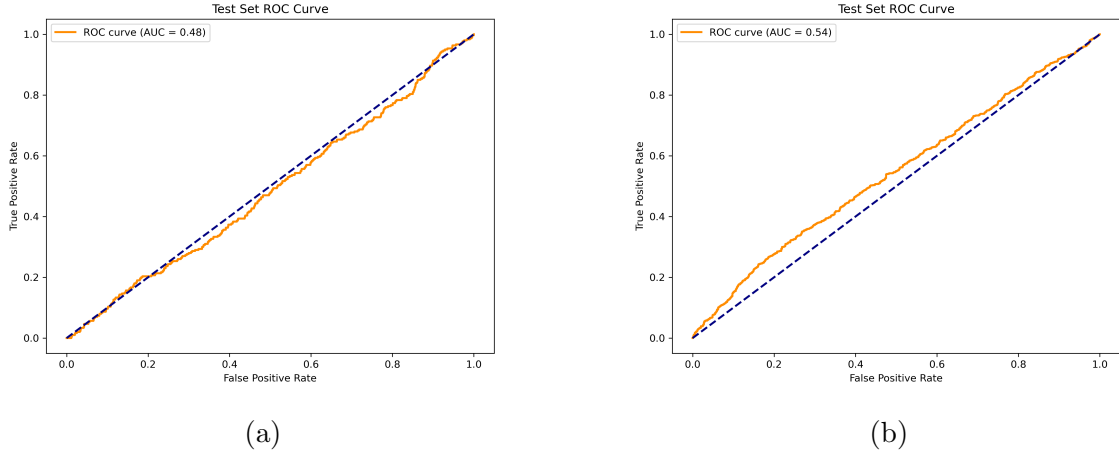


Figure 5: (a)AUC-ROC curve from the classification performed on the MAMOS data after training the MLP on ViT features from the CMMD dataset (b)AUC-ROC curve from the classification performed on the CMMD data after training the MLP on ViT features from the MAMOS dataset

Conclusion

In this dissertation we worked on testing the ability of deep learning based models to classify the images of varying resolution into categories based on their underlying pathology.

The major hypothesis being that the fourth power of resolution of images of radiological origin in terms of number of pixels is directly proportional to the amount of incident X-ray radiation, and when we strive to reduce the incident radiation while capturing an image, we lose on the resolution necessary for radiologists to analyze the images. Here, comes a deep learning based solution where we can extract features and then use neural network based classifiers to classify the images. In course of this study, we employed a CNN (EfficientNet-B0) and a ViT (MONAI ViT) to test the information content of the corresponding representations. We primarily used the “MAMOS_1000000004” dataset for doing all the feature extraction and trainings. Another, “CMMD” dataset was used to test the transferability of learning of the MLP.

Our primary results revealed that although the CNN like EfficientNet-B0 is very much capable at capturing a lot of important pathological details at varying resolutions, a domain specific ViT such as MOANI ViT, is much better than the CNN architecture. However, when we tried to check the cross-data transferability of both the features, the performance (in terms of AUC-ROC) essentially remained at random chance.

Future Directions

It is in our consideration that the results which we have got can be refined further, using better hyperparameter tuning and running the models for larger number of epochs, we can extrapolate the results on a larger scale, which will be better for a real-life scenario. The third objective of this project, which was exploring and applying transfer learning still requires further, rigorous experimentation and we are optimistic that this will be an important and interesting study to pursue.

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